

	Unit	Meaning		
<i>tokens</i>	Token	Input (sold) and output (bought) (i, o) is an ordered input-output pair and the direct route is $i \rightarrow o$. Generic token index used where a quantity applies to any endpoint or candidate token.		
k	Token	Candidate vehicle token used as the routes intermediate; $k \notin \{i, o\}$.		
h	Token	$h \in K \setminus \{i, o, k\}$.		
ℓ, p, p'	Token / pool	ℓ indexes tokens in cross-token routes and p indexes a direct liquidity pool; p' indexes another pool in leavest-out sums.		
$\hat{a}, \hat{a}'$	LP address	$\hat{a}$ indexes a local liquidity-provider address or resolved position controller; $\hat{a}'$ indexes another provider in within-pools.		
b	Fraction	Symmetric price-band half-width used to construct a symmetric liquidity around the current reference price; $b = 0.02$ is primary.		
t, u, w	UTC day / UTC week	t indexes a focal calendar day; u indexes another calendar day in a time-window sum; w indexes calendar weeks in UTC-weekly sums.		
τ	Days	Positive calendar-day horizons selected ex-ante for each dynamic specification.		
d, μ	Calendar days / calendar months	d is calendar-day event time relative to a candidate-specific shock; μ is calendar-month event time relative to an architecture activation date. Event time zero contains the event or activation date.		
g	Settlement comparison cell	Settlement comparison cell defined by ordered pair (i, o) , vehicle k , UTC week w , and a prospective route-link g .		
q	USD	Input quote notional. Un-audited route-cost columns use $q = \$10,000$.		
r	Route unit	Reconstructed route-unit cost per unit. A coherent $t \rightarrow k \rightarrow o$ component contributes one r regardless of its number of legs; a split or join contributes one r per reconstructed input-output pair.		
<i>Route and liquidity aggregates</i>	USD	$V_{0,t}$ is total realized USD volume across all direct and indirect route units on day t . $DVol_t$ restricts it to direct routes, so $0 \leq DVol_t \leq V_{0,t}$.		
$IVol_t, IVol_{k,t}$	USD	$IVol_t$ restricts $V_{0,t}$ to indirect routes, so $0 \leq IVol_t \leq V_{0,t}$. $IVol_{k,t}$ further restricts that volume to routes using vehicle k , so, also, $0 \leq IVol_{k,t} \leq IVol_t$.		
$Vol_{i,o,t}, IVol_{i,o,t}, IVol_{i,o,k,t}$	USD	$Vol_{i,o,t}$ is realized route-unit volume for ordered pair (i, o) on day t . $IVol_{i,o,t}$ restricts it to indirect routes, and $IVol_{i,o,k,t}$ further restricts it to routes through k , so $0 \leq Vol_{i,o,k,t} \leq IVol_{i,o,t} \leq Vol_{i,o,t}$.		
$N_t, N_{i,o,t}^{in}, N_{k,t}^{in}$	Route-unit count	N_t counts all reconstructed route-unit on day t ; not the individual legs. $N_{i,o,t}^{in}$ and $N_{k,t}^{in}$ restrict that count to routes with k as the input or output endpoint, respectively, each lies between 0 and N_t .		
$N_t^I, N_{i,o,t}^I, N_{k,t}^I$	Route-unit count	N_t^I restricts N_t to route units with at least one intermediate token. $N_{i,o,t}^I$ further restricts that count to units with intermediate k , so $0 \leq N_{i,o,t}^I \leq N_t^I \leq N_t$; superscript I denotes indirect routes.		
$A_t, A_{i,o,t}, A_{k,t}^I$	Sets of token pairs	A_t is the set of endpoint pairs selected in indirect routes on day t . $A_{i,o,t}^I$ restricts it to pairs using k , and $A_{k,t}^I$ further restricts it to pairs for which k has the largest candidate-vehicle volume; $A_{i,o,t}^I \subseteq A_{k,t}^I \subseteq A_t$.		
$LegVol_{i,o,t}^{in}, LegVol_{k,t}^{in}$	USD	$LegVol_{i,o,t}^{in}$ and $LegVol_{k,t}^{in}$ are route-leg USD volumes for which k is the leg's input or output token, respectively, both equal to q .		
$Vol_{i,o,t}^{in}, Vol_{k,t}^{in}$	USD	Input- and output-endpoint restrictions of $V_{0,t}$ for token k : $0 \leq Vol_{i,o,t}^{in} \leq V_{0,t}$ and $0 \leq Vol_{k,t}^{in} \leq V_{0,t}$.		
$K, K_{i,o,t}, K_{k,t}, m_p$	Set of tokens	Prespecified candidate set (WETH, USDC, USDT, DAI, WBTC).		
$\mathcal{L}_t, \mathcal{L}_{k,t}, m_p$	Set of pools / token count	$\mathcal{L}_t$ contains pools from protocols whose deposited-coupon contract is admitted, with exact token contracts and $0 < Capital_{k,t} \leq 10$ billion USD. $\mathcal{L}_{k,t} \subseteq \mathcal{L}_t$ restricts that set to pools with candidate k on one side. m_p is the number of pool tokens in K , so $m_p \in \{1, 2, 3\}$.		
$Capital_{p,t}, C_{k,t}$	USD	$Capital_{p,t}$ is pool p 's validated day- t accounting capital under its protocol contract. $C_{k,t} = \sum_{p \in \mathcal{L}_{k,t}} Capital_{p,t}/m_p$ is one-candidate pool contributes one half to each.		
$PoolVol_{p,t}, SpokenVol_{k,t}$	USD	$PoolVol_{p,t}$ is total realized swap volume in pool p on day t . $SpokenVol_{k,t}$ restricts it to pool legs belonging to reconstructed indirect routes that use k as the intermediate; $0 \leq SpokenVol_{k,t} \leq PoolVol_{p,t}$.		
<i>LP portfolio objects</i>	USD	End-of-day USD value of active liquidity supplied by provider a to pool p , marked at the day- t reference price $P_{k,t}$ realized by realized USD volume, where clean means route class <i>single</i> or <i>coherent</i> .		
$w_{a,p,t}$	Fraction (0-1)	Provider a 's share of active capital input p on day t .		
$R_{a,p,t}^{LP}$	USD per day	Day- t liquidity deposits by a into p minus withdrawals, valued at transaction-time USD prices and excluding fees paid by a .		
$Fees_{a,p,t}, G_{a,p,t}^{LP}$	USD	Swap fees accrued to provider a in pool p during day t , and USD gas spent by a on that position's mint, burn, collect, and rebalance transactions.		
$R_{a,p,t}^{LP}, R_{a,p,t}^{other}, R_{a,p,t}^{other}$	USD	End-of-day values of the day- t opening LP inventory and its self-financing rebalanced benchmark, respectively, both excluding day- t fees, flows, and gas.		
$R_{a,p,t}^{LP}, R_{a,p,t}^{other}, R_{a,p,t}^{other}$	Daily return fraction	$R_{a,p,t}^{LP} = (V_{a,p,t}^{LP} - L_{a,p,t-1})/L_{a,p,t-1}$ is the position's fee-, flow-, and gas-exchanged return. $R_{a,p,t}^{other}$ is the lagged-capital-weighted mean of that return across a 's pools other than p .		
$\omega_{a,t}, \omega_{a,t-p,t}$	Fraction (0-1)	Share of provider a 's shared LP inventory of post-V3 fixed t -exposed to token x , excluding both tokens in p and normalized to sum to one across the remaining tokens.		
$BandDepth_{p,t,k}$	USD	Average across trade directions of the input USD notional executable in pool p before its marginal price exits the symmetric b band around the day- t reference price.		
<i>Route and liquidity aggregates</i>	Daily log-return fraction	Day- t log return of the WETH price used to construct downside slopes.		
<i>Candidate-shock objects</i>	USD per token	$P_{k,t}$ is the day- t USD price of any token x . $P_{k,t}$ restricts that price to candidate $k \in K$.		
$P_{k,t}, R_{k,t}, R_{k,t}$	Daily log-return fraction	$R_{k,t} = \ln(P_{k,t}/P_{k,t-1})$ is the day- t return of any token x . $R_{k,t}$ restricts it to candidate k , and $N_{k,t}^{in}$ restricts it to post-V3 activity.		
$R_{k,t}^{(30)}, \sigma_{k,t}^{(30)}$	Daily log-return standard deviation	Sample standard deviation of $R_{k,t}$ over the 30 calendar days ending at $t = 1$, requiring at least 20 valid daily returns.		
<i>Architecture objects</i>	Daily log-return fraction / daily log-return standard deviation	$R_{k,t} = \ln[P_{k,t}/P_{k,t-1}]/(P_{k,t-1} - P_{k,t-2})$ is the ordered-endpoint-pair return. $\sigma_{k,t}^{(30)}$ is its sample standard deviation over the T_{pre}^{V3} days.		
<i>Persistence and displacement objects</i>	Token	$k_{i,o,t}^{in}$ is the incumbent vehicle with the largest mean $h_{i,o,t}^{in}$. $h_{i,o,t}^{in}$ is the executable nonincumbent candidate with the smallest all-in indirect cost $C_{i,o,k,t}^{in}$ on day t .		
<i>Architecture objects</i>	UTC day / set of UTC days / set of token pairs	t_{pre}^{V3} is the Uniswap V3 activation date, 5 May 2021. T_{pre}^{V3} is the fixed 180-calendar-day window ending at t_{pre}^{V3} . P_{pre}^{V3} is the fixed architecture-analysis universe of ordered pairs with positive realized route volume on at least 30 days in that pre-period; every member is quoted at q throughout the event window, independent of post-V3 activity.		
$t_{pre}^{V3}, T_{pre}^{V3}, P_{pre}^{V3}, t_{pre}^{V4}, T_{pre}^{V4}, P_{pre}^{V4}$	UTC day / sets of UTC days	t_{pre}^{V4} is the Ethereum V4 activation date established from the canonical deployment metadata before deposited-coupon contract is admitted, with exact token contracts and $0 < Capital_{k,t} \leq 10$ billion USD. $\mathcal{L}_{k,t} \subseteq \mathcal{L}_t$ restricts that set to pools with candidate k on one side. m_p is the number of pool tokens in K , so $m_p \in \{1, 2, 3\}$.		
<i>Route-cost quote objects</i>	Set of token pairs	$Vol_{i,o,t}$ is pool p 's validated day- t accounting capital under its protocol contract. $C_{k,t} = \sum_{p \in \mathcal{L}_{k,t}} Capital_{p,t}/m_p$ is one-candidate pool contributes one half to each.		
$P_{k,t}, q$	Set of token pairs	Common-support pairs for candidate k , day- t , and notional q : $C_{k,t} = P_{k,t} \cap \mathcal{L}_{k,t}$, so both the direct and indirect route are executable.		
$\mathcal{L}_{k,t}, \mathcal{L}_{k,t}$	Sets of token pairs	$\mathcal{L}_{k,t}$ is the third-direct subset. $W_{k,t} \subseteq \mathcal{L}_{k,t}$ contains common-support pairs for which the indirect route beats the direct route, equivalently $\Delta C_{i,o,k,t}^{in} < 0$.		
$\mathcal{L}_{k,t}, \mathcal{L}_{k,t}$	Sets of token pairs	Indicators for direct-route availability (D), indirect-route-through- k availability (I), and executable direct route return less than $0.5q$ (T).		
$\mathcal{L}_{k,t}, \mathcal{L}_{k,t}$	Sets of token pairs	Quoted output values, superscripts D and I denote direct and indirect routes.		
$\mathcal{L}_{k,t}, \mathcal{L}_{k,t}$	Sets of token pairs	Counterfactual direct and indirect route returns when this pool remains at its pre-quote marginal price but charges its historical swap fee; these retain fee loss and suppress price impact.		
$\mathcal{L}_{k,t}, \mathcal{L}_{k,t}$	Sets of token pairs	Historical gas expenditure for the direct and indirect routes, respectively, using route-specific gas usage and the day- t gas-token USD price.		
$\mathcal{L}_{k,t}, \mathcal{L}_{k,t}$	Sets of token pairs	All-in execution-cost fractions for <i>direct</i> and <i>indirect</i> plus historical gas expenditure.		
$\mathcal{L}_{k,t}, \mathcal{L}_{k,t}$	Sets of token pairs	Pair-level direct cost advantage $(O_{i,o,t}^{in} - O_{i,o,k,t}^{in})/O_{i,o,t}^{in}$ on $C_{k,t}$; $C_{k,t}$ positive values favor the direct route.		
$\mathcal{L}_{k,t}, \mathcal{L}_{k,t}$	Sets of token pairs	Pair-level indirect cost advantage $(C_{i,o,k,t}^{in} - C_{i,o,t}^{in})/C_{i,o,t}^{in}$ on $C_{k,t}$; $C_{k,t}$ positive values favor the indirect route.		
<i>Settlement objects and operators</i>	Set of route units	Receipt-audited matched route units using vehicle k in UTC week w .		
$R_{k,w}, R_{k,w}^{in}, R_{k,w}^{in}$	Set of route units	Members whose receipt legs a transfer of vehicle k : $R_{k,w}^{in} = R_{k,w}^{in}$.		
$R_{k,w}, R_{k,w}^{in}, R_{k,w}^{in}$	Set of route units	Receipt-audited route units in settlement comparison cell g executed on Uniswap V3 and V4, respectively. A matched cell has both sets nonempty.		
$M_{k,t}, GrossLegVol_{k,t}$	USD	$M_{k,t}$ is the sum of absolute, route-attributed ERC-20 transfers of intermediate k between distinct nonzero addresses in route unit r , valued in USD. $GrossLegVol_{k,t}$ is the corresponding gross USD notional of k entering or leaving the route's pools across both vehicle legs.		
$ \mathcal{S} , \mathbf{1}_{\{\cdot\}}$	Count / indicator (0/1)	Cardinality of any finite set $\mathcal{S}$ and an indicator equal to one when its subscripted condition is true.		
<i>Dynamic operators</i>				
Δ, Δ_t		Change in a daily variable over the τ days ending at t : $\Delta_t = X_t - X_{t-\tau}$.		
Variable	Formula	Unit	Data column	Definition
<i>Vehicle-use measures</i>			bridge.share	Fraction of days of indirect-route volume routed through k .
$VehicleShare_{k,t}$	$\frac{IVol_{k,t}}{IVol_t}$	Fraction (0-1)	all.route.bridge.share	Fraction of days all-route USD volume routed indirectly.
$AllRouteVehicleShare_{k,t}$	$\frac{IVol_{k,t}}{V_{0,t}}$	Fraction (0-1)	intermediate.share	Token k 's share of clean, non-cyclical value carried as an intermediary; k include one reconstructed component is counted once. Primary shares use the prespecified currency types and exclude the residual unclassified contract bucket.
$IShare_{k,t}$	$\frac{IVol_{k,t}}{\sum_{j \in \mathcal{K} \setminus \{i,o\}} Type(j) \cdot other IVol_{j,t}}$	Fraction (0-1)	endpoint.share	Token k 's share of source-or-sink value across all clean, non-cyclical routes, including direct routes where k is the intermediary; one component's appearance is counted once.
$EShare_{k,t}$	$\frac{EVol_{k,t}}{\sum_{j \in \mathcal{K} \setminus \{i,o\}} Type(j) \cdot other EVol_{j,t}}$	Fraction (0-1)	intermediate.share	Token k 's share of clean, non-cyclical route components in which it is an intermediary; one component's appearance is counted once.
$ExcessUse_{k,t}$	$\frac{IShare_{k,t}}{IShare_{k,t}^{in}}$	Ratio	vehicle.excess.use.ratio	Vehicle k 's share of source-or-sink value across all clean, non-cyclical routes, including direct routes where k is the intermediary; one component's appearance is counted once.
$IShare_{k,t}^{in}$	$\frac{N_{k,t}^{in}}{\sum_{j \in \mathcal{K} \setminus \{i,o\}} Type(j) \cdot other N_{j,t}^{in}}$	Fraction (0-1)	intermediate.share	Token k 's share of clean, non-cyclical route components in which it is an intermediary; one component's appearance is counted once.
$EShare_{k,t}^{in}$	$\frac{N_{k,t}^{in}}{\sum_{j \in \mathcal{K} \setminus \{i,o\}} Type(j) \cdot other N_{j,t}^{in}}$	Fraction (0-1)	endpoint.share	Token k 's share of source-or-sink value across all clean, non-cyclical routes, including direct routes where k is the intermediary; one component's appearance is counted once.
$ExcessUse_{k,t}^{in}$	$\frac{IShare_{k,t}^{in}}{IShare_{k,t}^{in}}$	Ratio	vehicle.excess.use.ratio	Vehicle k 's share of source-or-sink value across all clean, non-cyclical routes, including direct routes where k is the intermediary; one component's appearance is counted once.
$IShare_{k,t}^{in,20}$	$\frac{N_{k,t}^{in,20}}{\sum_{j \in \mathcal{K} \setminus \{i,o\}} Type(j) \cdot other N_{j,t}^{in,20}}$	Fraction (0-1)	within.20pct	Count share on the exact route components used by the strict-value measure: source, sink and every intermediary reconcile within 20 percent.
$EShare_{k,t}^{in,20}$	$\frac{N_{k,t}^{in,20}}{\sum_{j \in \mathcal{K} \setminus \{i,o\}} Type(j) \cdot other N_{j,t}^{in,20}}$	Fraction (0-1)	endpoint.share.within.20pct	Endpoint count share on the same 20-percent reconcile value-coherence support.
$ExcessUse_{k,t}^{in,20}$	$\frac{IShare_{k,t}^{in,20}}{IShare_{k,t}^{in,20}}$	Ratio	vehicle.excess.use.ratio.within.20pct	Matched-support count excess use; undefined when strict-support endpoint count is zero.
$IShare_{k,t}^{in,20}$	$\frac{IVol_{k,t}^{in,20}}{\sum_{j \in \mathcal{K} \setminus \{i,o\}} Type(j) \cdot other IVol_{j,t}^{in,20}}$	Fraction (0-1)	intermediate.share.within.20pct	Intermediary value share where source, sink and every intermediary reconcile within 20 percent.
$EShare_{k,t}^{in,20}$	$\frac{EVol_{k,t}^{in,20}}{\sum_{j \in \mathcal{K} \setminus \{i,o\}} Type(j) \cdot other EVol_{j,t}^{in,20}}$	Fraction (0-1)	endpoint.share.within.20pct	Endpoint value share on the same 20-percent value-coherence support.
$ExcessUse_{k,t}^{in,20}$	$\frac{IShare_{k,t}^{in,20}}{IShare_{k,t}^{in,20}}$	Ratio	vehicle.excess.use.ratio.within.20pct	Strict-value intermediary share divided by strict-value endpoint share on identical support.
$VehicleCountShare_{k,t}$	$\frac{N_{k,t}^{in}}{N_t^{in}}$	Fraction (0-1)	bridge.count.share	Fraction of day- t indirect route units that use k .
$PairCoverage_{p,k,t}$	$\frac{ \mathcal{A}_t }{ \mathcal{A}_t }$	Fraction (0-1)	pair.coverage	Fraction of active indirect-route endpoint pairs for which k has the largest candidate-vehicle volume.
$MainVehiclePairShare_{k,t}$	$\frac{ \mathcal{A}_t^I }{ \mathcal{A}_t^I }$	Fraction (0-1)	pair.main.vehicle.share	Fraction of active indirect-route endpoint pairs for which k has the largest candidate-vehicle volume.
$IVol_{k,t}$	USD	USD	bridge.volume.usd	Sum of realized USD volume over indirect routes using vehicle k on day t .
<i>Pair-level route-choice measures</i>			actual.vehicle.share	Candidate k 's share of realized indirect-route USD volume for ordered pair (i, o) on day t , defined when $IVol_{k,t} > 0$.
$VehicleShare_{i,o,k,t}$	$\frac{IVol_{i,o,k,t}}{IVol_{i,o,t}}$	Fraction (0-1)	pair.indirect.route.share	Fraction of realized route-unit USD volume for ordered pair (i, o) executed through an indirect route on day t , defined inputs $Vol_{i,o,k,t} > 0$.
$IndirectRouteShare_{i,o,k,t}$	$\frac{IVol_{i,o,k,t}}{Vol_{i,o,t}}$	Fraction (0-1)	pair.indirect.route.share	Fraction of realized route-unit USD volume for ordered pair (i, o) executed through an indirect route on day t , defined inputs $Vol_{i,o,k,t} > 0$.
$Coverage_{k,t}^{in}$	$\frac{N_{k,t}^{in}}{N_t^{in} - N_{k,t}^{in} - N_{k,t}^{in}}$	Fraction (0-1)	pair.coverage.vehicle.coverage	Share of pair (i, o) 's realized indirect-route volume routed through k as the intermediary; one component's appearance is counted once.
$VehicleHHI_{i,o,t}$	$\frac{\sum_{k \in \mathcal{K} \setminus \{i,o\}} \left(\frac{VehicleShare_{i,o,k,t}}{Coverage_{k,t}^{in}} \right)^2}{\sum_{k \in \mathcal{K} \setminus \{i,o\}} \left(\frac{VehicleShare_{i,o,k,t}}{Coverage_{k,t}^{in}} \right)^2}$	Fraction (0-1)	pair.vehicle.hhi	Herfindahl concentration of indirect route units on day t , over re-normalizing their shares to sum to one; defined when candidate-routed indirect volume is positive.
<i>Network and route-denominator controls</i>			vol.share	$LegVol_{i,o,t}^{in}$ and $LegVol_{k,t}^{in}$ are input-side and output-side USD volumes for k ; ℓ indexes a candidate token in the day- t network.
$VolShare_{k,t}$	$\frac{LegVol_{i,o,t}^{in} + LegVol_{k,t}^{in}}{\sum_{\ell \in \mathcal{K} \setminus \{i,o\}} (LegVol_{i,o,\ell}^{in} + LegVol_{\ell,k,t}^{in})}$	Fraction (0-1)	betweenness centrality	Fraction of day- t intent routes on which k is intermediate, excluding output routes where k is the input or output endpoint. Superscripts identify each route role.
$RouteIncidence_{k,t}^{in}$	$\frac{N_{k,t}^{in}}{N_t^{in} - N_{k,t}^{in} - N_{k,t}^{in}}$	Fraction (0-1)	volume.weighted.betweenness	USD-volume analogue of RouteIncidence $_{k,t}^{in}$, using the same route-unit counts and input/output-endpoint exclusions.
$RouteIncidence_{k,t}^{in}$	$\frac{IVol_{k,t}}{Vol_t - Vol_{k,t}^{in} - Vol_{k,t}^{in}}$	Fraction (0-1)	daily.all.route.volume.usd	Total realized USD volume across direct and indirect route units on day t .
Vol_t	$DVol_t + IVol_t$	USD	daily.indirect.route.volume.usd	Total realized USD volume across indirect route units on day t .
$IVol_t$	$IVol_t$	USD	indirect.route.share	Fraction of days all-route USD volume executed through indirect routes.
$IndirectRouteShare_t$	$\frac{IVol_t}{Vol_t}$	Fraction (0-1)	vehicle.linked.capital.usd	Candidate k 's allocated share of validated deposited capital contributed by legs of indirect routes using k as the intermediate; defined for positive pool volume.
<i>LP capital measures</i>			lp.capital.share	Candidate k 's share of total candidate-linked deposited capital on day t .
$C_{k,t}$	$\sum_{p \in \mathcal{L}_{k,t}} \frac{Capital_{p,t}}{m_p}$	USD	log.vehicle.linked.capital	Natural log of one plus $C_{k,t}$, measured in USD.
$LPCapitalShare_{k,t}$	$\frac{C_{k,t}}{\sum_{\ell \in K} C_{\ell,t}}$	Fraction (0-1)	vehicle.capital.factor.100	Leave-one-out mean daily log deposited-capital change among other valid pools linked to candidate k ; requires at least three other pools.
$LogVehicleCapital_{k,t}$	$\ln(1 + C_{k,t})$	Natural-log points	market.capital.factor.100	Leave-one-out mean daily log deposited-capital change among all other valid pools in $\mathcal{L}_t$; minute sample, abundance is unsupported and never means the rob was initiated.
<i>LP capital commonality measures</i>			pair.direct.quote.quality	Direct-route output ratio for cells with $D_{i,o,k,t} = 1$ and $I_{i,o,k,t} = 0$; undefined when the direct route is not executable.
$VehicleCapitalFactor_{p,k,t}$	$\frac{\sum_{p' \in \mathcal{L}_{k,t}} \Delta_t \ln(Capital_{p',t})}{\sum_{p' \in \mathcal{L}_{k,t}} \Delta_t \ln(Capital_{p',t})}$	Daily log-change points	pair.indirect.quote.quality	Indirect-route output ratio through candidate k for $(i, o) \in \mathcal{L}_{k,t}$ undefined when that route is not executable.
$MarketCapitalFactor_{p,t}$	$\frac{\sum_{p' \in \mathcal{L}_{k,t}} \Delta_t \ln(Capital_{p',t})}{ \mathcal{L}_{k,t} \cdot (p)}$	Daily log-change points	pair.direct.fee.cost	Direct-route loss attributable to historical pool fees; price-impact measure is suppressed at the pre-quote marginal price.
<i>Liquidity commonality measures</i>			pair.direct.impact.cost	Additional direct-route output loss caused by moving the through the realized liquidity curve, after holding the fee-only output fixed.
$VehicleRouteShare_{p,k,t}$	$\frac{PoolVol_{p,k,t}}{PoolVol_{k,t}}$	Fraction (0-1)	pair.indirect.fee.cost	Two-fold indirect-route loss attributable to both pools' historical fees when price impact is suppressed at each pre-quote marginal price.
<i>LP capital and return measures</i>			pair.indirect.impact.cost	Additional indirect-route output loss caused by moving the through both realized liquidity curves, after holding fee-only output fixed.
$L_{a,p,t}$	USD	USD	pair.all.in.direct.cost	Historical direct-route gas expenditure as a fraction of input USD notional.
$w_{a,p,t}$	$\frac{L_{a,p,t}}{\sum_{a' \in \mathcal{K}} L_{a',p,t}}$	Fraction (0-1)	pair.all.in.indirect.cost	Indirect output-loss across both legs, which includes both pools' fees and price impact, plus historical route gas as a fraction of input notional. Fees, price-impact, and gas contributions are retained separately and sum to this measure.
$R_{a,p,t}^{LP}$	USD per day	USD per day	pair.all.in.direct.cost. advantage	Indirect all-in cost minus direct all-in cost on common support; positive values favor direct execution and negative values favor indirect execution.
$LPFeeYield_{a,p,t}$	$\frac{Fees_{a,p,t}}{L_{a,p,t-1}}$	Daily fraction	direct.available.share	Fraction of pairs in $\mathcal{P}_{k,t}$ with $D_{i,o,k,t} = 1$ and $I_{i,o,k,t} = 0$; equals one when the direct route is executable.
$LVR_{a,p,t}$	$\frac{V_{a,p,t}^{LP} - V_{a,p,t-1}^{LP}}{L_{a,p,t-1}}$	Daily fraction	vehicle.available.share	Fraction of pairs in $\mathcal{P}_{k,t}$ with $D_{i,o,k,t} = 1$ and $I_{i,o,k,t} = 0$; equals one when the direct route is executable.
$LPNetReturn_{a,p,t}$	$\frac{LPFeeYield_{a,p,t} - LVR_{a,p,t}}{G_{a,p,t-1}^{LP}}$	Daily fraction	no.direct.vehicle.available.share	Fraction of pairs with $D_{i,o,k,t} = 0$ and $I_{i,o,k,t} = 1$; no executable direct route but an executable route through k .
$R_{a,p,t}^{other}$	$\frac{\sum_{p' \neq p} L_{a,p',t-1} - R_{a,p,t-1}^{LP}}{\sum_{p' \neq p} L_{a,p',t-1}}$	Daily return fraction	direct.quote.quality.median	Median direct-route output ratio $O_{i,o,k,t}^{in}/q$ over $\mathcal{L}_{k,t}$ with $D_{i,o,k,t} = 1$ and $I_{i,o,k,t} = 0$; equals one when the direct route is executable.
$R_{a,p,t}^{other}$	$\frac{\sum_{p' \neq p} L_{a,p',t-1} - R_{a,p,t-1}^{LP}}{\sum_{p' \neq p} L_{a,p',t-1}}$	Daily return fraction	direct.cost. advantage.median	Median $\Delta C_{i,o,k,t}^{in}$ over common-support pairs; positive values favor the direct route and negative values favor the indirect route through k .
$R_{a,p,t}^{other}$	$\frac{\sum_{p' \neq p} L_{a,p',t-1} - R_{a,p,t-1}^{LP}}{\sum_{p' \neq p} L_{a,p',t-1}}$	Daily return fraction	all.in.direct.cost. advantage. median	Median pair-level all-in direct route cost advantage over common-support pairs; positive values favor direct execution after gas.
$LPWealthShock_{k,t}$	$\frac{\sum_n w_{a,p,t-1} - Z_{a,p,t-1}^{other}}{n}$	Daily return fraction	vehicle.beats.direct.shr	Fraction of $(i, o) \in \mathcal{L}_{k,t}$ for which $\Delta C_{i,o,k,t}^{in} < 0$.
$LPCapitalShare_{k,t}$	$\frac{C_{k,t}}{\sum_{\ell \in K} C_{\ell,t}}$	Fraction (0-1)	this.direct.share	Fraction of pairs with an ex-ante direct quote but $O_{i,o,k,t}^{in}/q < 0.5$. This is a quote-quality proxy, not a direct measure of pool liquidity depth.
$LogVehicleCapital_{k,t}$	$\ln(1 + C_{k,t})$	Natural-log points	post.v3	Equals one on and after the Uniswap V3 activation date.
<i>Routing-search efficiency measures</i>			v4.route	Equals one when matched route unit r executes on Uniswap V4.
$ChosenError_t$	$10^4 (O_{i,o,t}^{in} - O_{i,o,t}^{in}) / O_{i,o,t}^{in}$	Basis points	v4.route.share	Fraction of pairs with $V_{i,o,k,t} = 1$ and $I_{i,o,k,t} = 0$; equals one when the direct route is executable.
$ ChosenError_t^{max} $	$\max\{ e_1^{(1)} , e_2^{(2)} , e_3^{(3)} \}$	Basis points	pre.v3.direct.constraint	Fraction of the fixed 180-day pre-V3 window in which pool p lacks an executable direct route at notional q ; higher values mean more frequent direct route failures.
$Regret_t^{reach}$	$10^4 [\max(O_{i,o,t}^{in}, O_{i,o,t}^{in}) - \min(O_{i,o,t}^{in}, O_{i,o,t}^{in})] / O_{i,o,t}^{in}$	Basis points	pre.v3.pair.volatility	Sample standard deviation of ordered endpoint-pair log return over the fixed 180-day pre-V3 window; computed only from information available before V3 activation.
$ReachIncrement_t$	$10^4 [\max(O_{i,o,t}^{in}, O_{i,o,t}^{in}) - \max(O_{i,o,t}^{in}, O_{i,o,t}^{in})] / O_{i,o,t}^{in}$	Basis points	pool.fee.depth.capital. efficiency	Fee-inclusive executable dollar depth, measured by the fixed 180-day pre-V3 window in which pool p lacks an executable direct route at notional q ; higher values mean more frequent direct route failures.
$PathChoiceIncrement_t$	$10^4 [\max(O_{i,o,t}^{in}, O_{i,o,t}^{in}) - \max(O_{i,o,t}^{in}, O_{i,o,t}^{in})] / O_{i,o,t}^{in}$			